

Grade 1 Science Unit 4 – Review

Revise the ideas and concepts taught in:

Unit 4 - Humans and Animals Grow

- Talk to your child about what they remember and the definitions of the key vocabulary.
- Ask them to explain the ideas to you.
- Answer the questions in the book.
- Complete the experiments and enquiries.

4 Humans and animals grow

> 4.1 Comparing animals

We are going to:

- find out how animals look similar and different
- ask our own questions about animals
- use books, videos or the internet to answer questions
- use and make block graphs.

Getting started

- Look at these animals. How are they different? How are they similar?
- Talk about your ideas with your class.



bird fur mammal
feathers human



4.1 Comparing animals

Compare these animals.

The giraffe looks different from the zebra because it has a longer neck.

The giraffe looks similar to the zebra because they both have four legs and a tail.

Giraffes and zebras both have **fur**. Fur is similar to hair.

Animals that have hair or fur are called **mammals**.

People are a special kind of mammal.

We are called **humans**.

Zebra fur



Giraffe fur



Human hair



Birds look different to mammals.
They do not have fur or hair.

Birds have **feathers**.

A bird looks different from a human in many other ways.

A bird looks similar to a human because they both have two legs.

Questions

- 1 Find another way the giraffe looks different from the zebra.
- 2 Find two other ways the giraffe and zebra look similar.
- 3 How does a bird look different from a human?
Think of as many ways as you can.



Think like a scientist 1

Asking questions about animals

You will need:
some books about animals and a computer

Marcus has a question about animals.
How can Marcus find out the answer?

Marcus does not have a real elephant to observe so he uses a book, a video or the internet.

He finds out that elephants do have hair.
Elephants are mammals like humans.

Write your own questions about animals.

Use a book, a video or the internet to find the answers.



Observing small animals

Sofia has another question about animals.
How can Sofia find out the answer?

How many legs
do very small animals
have?



Sofia looks for some very small animals and observes them.



Sofia makes a block graph to show her results.

Questions

- How many animals did Sofia find with no legs?
- How many animals did she find with 6 legs?
- How many animals did she find in total?

Think like a scientist 2

How many legs?

You will need:
a magnifying glass

Continued

Find an outdoor habitat in your school.
Observe the small animals that live there.
Use a magnifying glass to make them easier to see.
How many legs do they have?

Do not touch animals that can sting or bite.

Make your own block graph to show your results.

How am I doing?

Look at one of the small animals with a friend.
If you count a different number of legs, count again to check.

Asking questions is an important part of being a scientist.

Do you ask questions about the world?

Does asking questions help you to learn?

You could ask a question that helps people find out something new.

Are there any animals with 5 legs?

Look what I can do!

- ☐ I can say how animals look similar and different.
- ☐ I can ask my own questions about animals.
- ☐ I can use a book, video or the internet to answer questions.
- ☐ I can use and make a block graph.

> 4.2 Growing

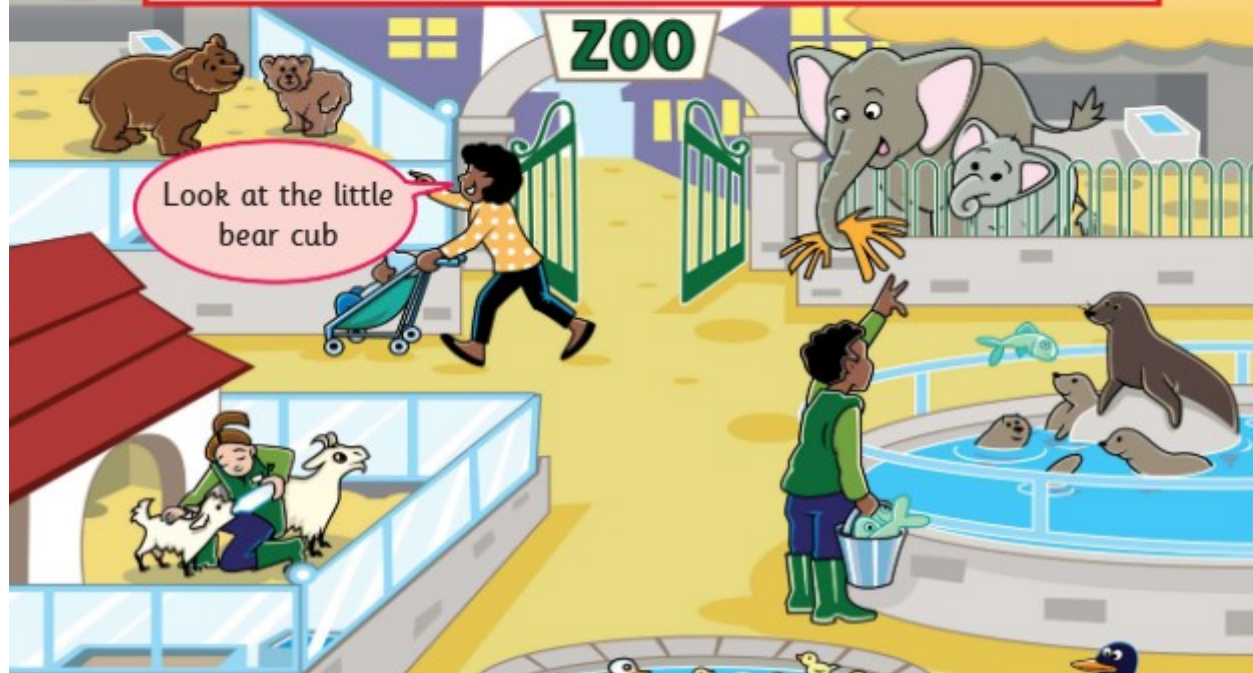
We are going to:

- learn how humans and other animals change as they grow
- measure how tall we are using bottles
- draw block graphs
- find patterns in results.



Getting started

- How many types of **baby** animal can you see?
- Do the babies look similar to their **parents**?
- Which baby animals look similar to their parents? Tell a friend.
- Now say which baby animals look different to their parents.



Adults and babies

All animals make babies. The babies are called the animals' **young**.

The young change as they grow **older**. This is called **growth**.

When they are fully grown, we call the animals **adults**.

The young of many animals look similar to the adults.

But the young of some animals look different from the adults.

A young elephant is called a **calf**.

When it gets older it will grow bigger.

adult	growth/grow
age	old/older
baby	parent
calf	pictogram
chick	young
cub	



A young penguin is called a **chick**.

When it gets older it will grow different coloured feathers.

Questions

- 1 The calf looks very similar to the adult elephant. How is it different?
- 2 The chick looks very different from the adults. How is it similar?



Activity 1

How do baby animals change?

Marcus has a picture of a bear and a baby bear. We call a baby bear a **cub**. He adds notes to the picture to show how the cub changes as it grows.

Look at some pictures of baby animals and adult animals to see how baby animals change as they grow.

Find or draw a picture of an animal and its young.

Make notes to show how the baby animal changes as it grows.



Here is a human baby.



Question

- 3 How do humans change as they grow?

Activity 2

A younger you

Sofia has found a photo taken when she was a baby.

Look at a photo of yourself when you were younger. What did you look like then?

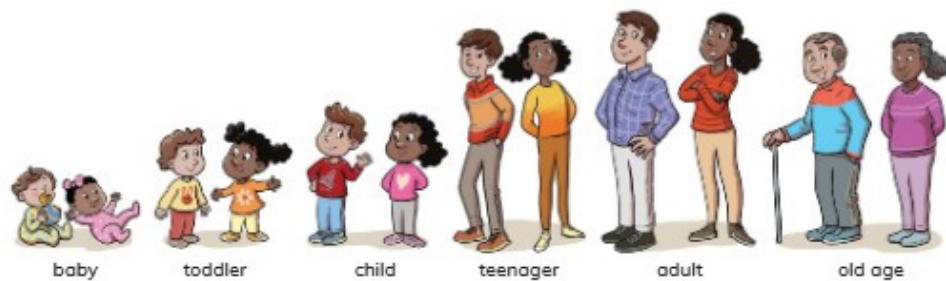
How old are you now? This is called your **age**.

Compare the photo of yourself when you were younger with how you look now. Talk about how you have changed.



How humans grow

These pictures show how humans change as they grow older.



Question

4 Which picture shows people the same age as you?

Sofia uses bottles to measure the height of some people of different ages.



She makes a **pictogram** to show the results.

A pictogram is like a block graph but uses pictures instead of blocks.

Look at the ages and the number of bottles. What is the pattern?

Questions

- 5 How many bottles tall was the child aged 10?
- 6 How many bottles tall was the adult aged 30?
- 7 Describe the pattern in the results.

Numbers of bottles tall	7			
	6			
	5			
	4			
	3			
	2			
	1			
		Toddler aged 5	Child aged 10	Adult aged 30



Think like a scientist

Measuring height

You will need:

two identical plastic bottles, a younger child and an adult

Use plastic bottles to measure yourself, a younger child and an adult.



Make a block graph to show your results.

How am I doing?

Ask a friend to check your measuring.

Have you counted the right number of bottles?

Look what I can do!

- ☐ I can describe how humans and some animals change as they grow.
- ☐ I can measure how tall a person is using bottles.
- ☐ I can draw a block graph.
- ☐ I can find patterns in results.

> 4.3 Inheriting characteristics

We are going to:

- find out why animals look similar to their parents
- put children into groups using characteristics.

Getting started

- These animals are all sheep. They all look different.
- Why do you think they look different? Tell a friend what you think.



The sheep look different because they have different characteristics.

Some have long wool. Some have short wool.

Some have black faces. Some have white faces.

The sheep are different colours.

The sheep have different characteristics because they have different parents. All animals get some of their characteristics from their parents.

We say that animals **inherit** characteristics from their parents.

Animals look similar to their parents but they do not look the same.

We say they are not **identical**. This is because an animal inherits some characteristics from one parent and some characteristics from the other parent.

Every tiger has a different pattern of **stripes**.

A tiger cub's stripes are not the same as its parents' stripes, but they are similar.



This is a human **fingerprint**.

Every person has a different pattern in their fingerprint.

Your fingerprints are not the same as your parents' fingerprints, but they are similar.

There are many other characteristics we inherit from our parents.

Usually they make us look different from each other.

These children are identical **twins**. They look the same because they have inherited the same characteristics from their parents.



Activity

How are you different?

Here are five characteristics.

Which characteristics do you have? Look in a mirror.

 Freckles	  Free Fixed	  Left handed Right handed
 Tongue rolling		  straight hair curly hair

Continued

Talk to your friends.

Make groups of children with similar characteristics.

Does anyone have all the same characteristics as you?

Think like a scientist

Sorting fingerprints

You will need:
a balloon and a marker pen

Observe your fingerprints to find out which type of fingerprint you have.

Look at the pictures to see what to do.



Continued

Compare your fingerprint with these three pictures.
Which types of fingerprint do you have?



loop



whorl



arch

How am I doing?

Have a look at your friends' fingerprints. Which types do they have?
Have they sorted their fingerprint correctly?

Fingerprint patterns are difficult to see. You used a balloon to make them larger. Can you think of a different way of looking at fingerprints? What could you use to make fingerprints look larger?

Look what I can do!

- ☐ I can explain why animals look similar to their parents.
- ☐ I can put children into groups using characteristics.

> 4.4 Keeping healthy

We are going to:

- find out how to keep healthy
- find out why humans get ill and what it does to our bodies
- sort food into groups
- say if our predictions are correct or not.

Getting started

- Bimla wants to eat food that is good for her. What should she eat?

Eat chocolate. Sugar gives you energy.

Eat food that tastes good.

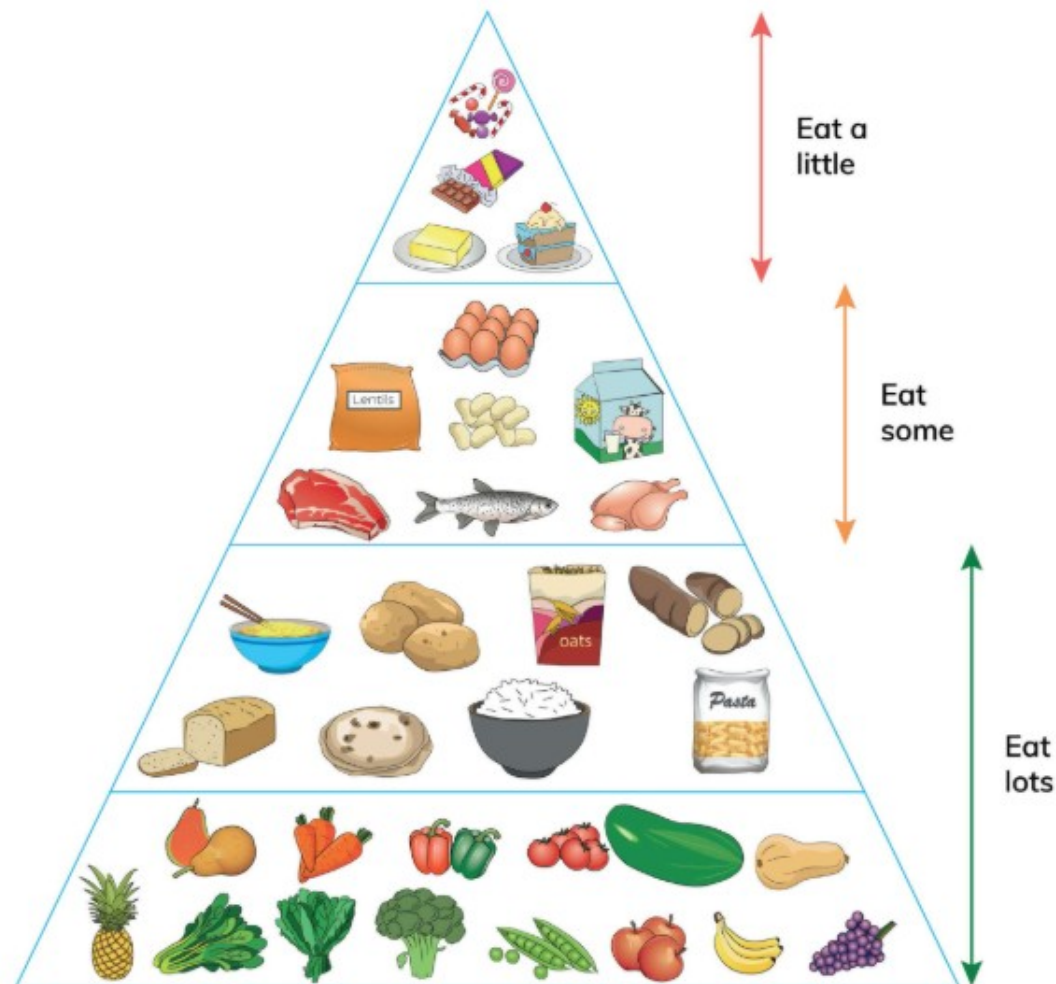
Drink milk. It makes you strong.

Fruit and vegetables are good for you.

clean	germs	ill	sick
diet	headache	illness	sweat
exercise	heart	muscles	

Healthy eating

Food that is good for us is called healthy food. Humans need to eat lots of different foods to stay healthy. The food we eat is called our **diet**.



This food triangle shows how much of each type of food we should eat. If we do not have a healthy diet our bodies might not get what they need. This is one way we can become **ill**.

Activity

Sorting food

Marcus and Zara are sorting food into groups.

Try doing this with some foods or pictures of foods. Look at the food triangle to work out where to put each food.



Being ill

Nobody feels good all the time. When we feel **sick** or have a **headache** we say we are ill or we have an **illness**.

Germs are living things that are too small to see. Some germs can make us ill. This happens when the germs get inside us and our bodies cannot kill them. This is another way we can become ill.



Do these things to stop germs getting into your body.

Wash your hands after going to the toilet.

Wash your hands before eating.

Do not eat dirty food or drink dirty water.



Sometimes germs can make us cough or sneeze.

Cover your mouth when you cough or sneeze.

This can stop germs getting to other people.

Healthy people become ill less often.

Their bodies are better at killing germs.

Question

1 Have you had an illness? How did you feel?



Exercise and keeping clean

We can also make our bodies healthy by doing **exercise**. Exercise is when we use our **muscles** to move our bodies. It makes our muscles stronger.

Humans should do some exercise most days.

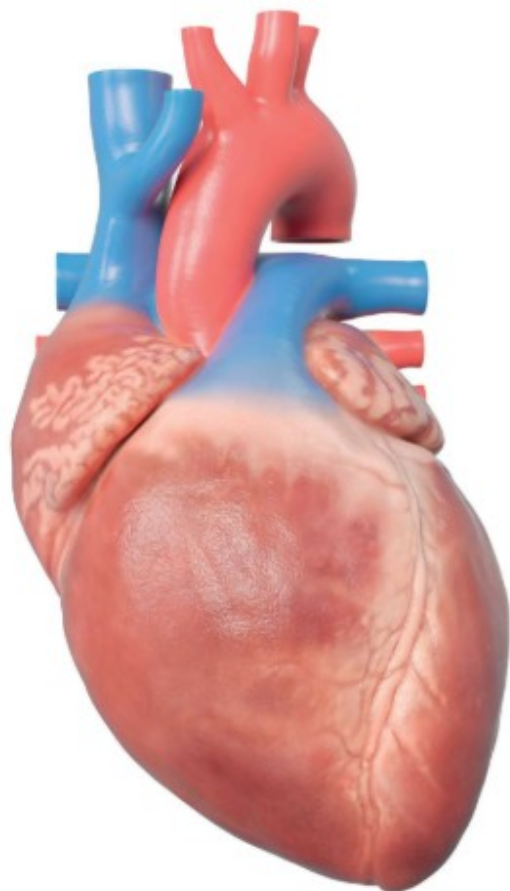
Your **heart** is a muscle.

Exercise is good for your heart.

Exercising makes us **sweat**.

We need to wash after exercising to keep **clean**.

Keeping clean is another way to keep your body healthy.



Questions

2 Which of these is exercise?

running

reading

swimming

playing football

sleeping

3 Which parts of the body get stronger when you exercise?

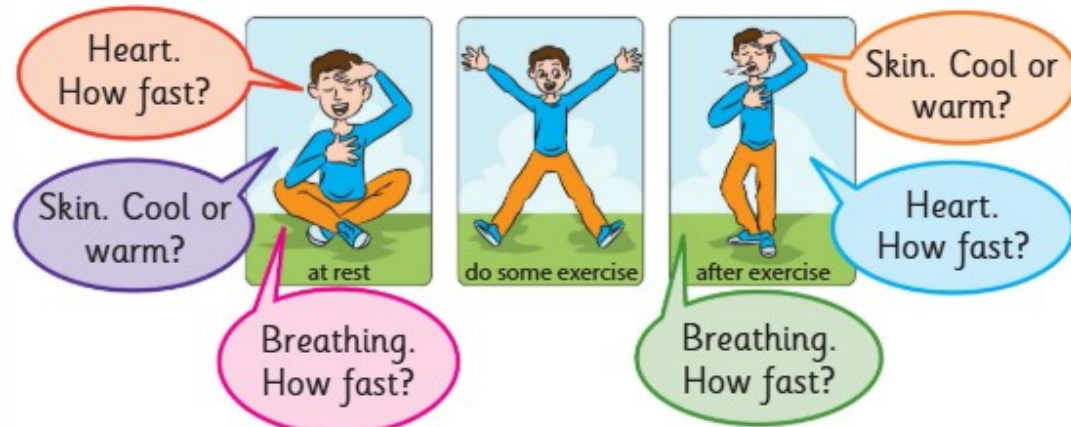
4 Why is it good to wash your hands before eating?



Think like a scientist

What happens when we exercise?

You will need: a timer



Look at the pictures to see what to do.

Tell a friend what you think will happen.

Think about your breathing, your heart and your skin.

Were your predictions correct? Did you observe any other changes?

How am I doing?

Were your friend's predictions correct?

Tell them what you think.

Look what I can do!

- ☐ I can talk about three ways to keep healthy.
- ☐ I can say why humans get ill and what being ill does to our bodies.
- ☐ I can sort food into groups.
- ☐ I can say if my predictions are correct or not.

> 4.5 Teeth

We are going to:

- learn about the different types of teeth and what they do
- learn about how to look after teeth
- say if our predictions are correct or not
- observe human teeth and make models of them
- investigate teeth safely.

Getting started

- Draw some teeth.
- Add labels to show what you know about teeth.

calcium	gums
canine	incisor
dairy	molar
dentist	toothpaste
fluoride	

Looking after teeth

We need to look after our teeth.

We use teeth to eat and we also use teeth to talk.

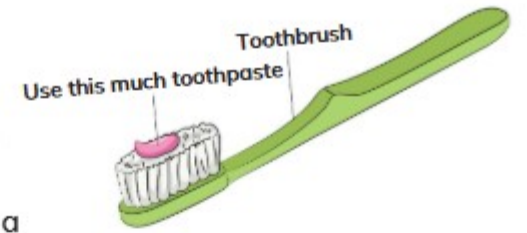
You can keep your teeth healthy by eating a healthy diet. **Calcium** is a material that helps to make teeth strong. Milk and other **dairy** foods, some beans and some fish have calcium in them. Make sure you eat some of these foods.

Food with lots of sugar is bad for teeth. Eat very little food with lots of sugar.

Brush your teeth twice every day for two minutes to keep them clean. Use **toothpaste** with **fluoride** in it. Fluoride helps to make your teeth strong.

Do not use too much toothpaste.

Go to the **dentist** twice a year. The dentist will check your teeth are healthy and can help you look after them.



Questions

- 1 What three things can you do to look after your teeth?
- 2 How much toothpaste should you use?
- 3 What do calcium and fluoride do?

Think like a scientist 1

What does sugar do to teeth?

You will need:

a drink with lots of sugar, water, egg shells and two cups

Teeth are made of a similar material to egg shells.

To find out what sugar does to our teeth we can investigate what sugar does to egg shells. Look at the pictures to see what to do.

Predict what will happen to the egg shell in each cup.

Record your prediction in a diagram.



Place half an egg shell into each cup.



Add water to one cup and a sugary drink to the other.



After seven days see what has happened to the egg shell.

What did you find out at the end?

Explain what too much sugar can do to teeth.

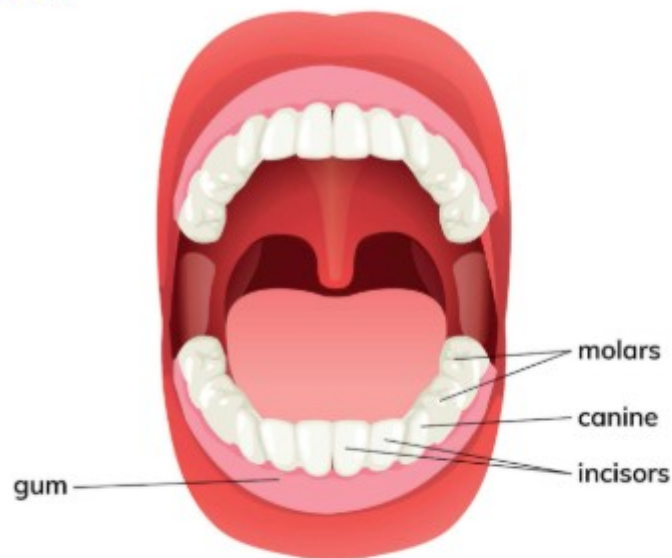
How am I doing?

Look at your predictions.

Can you say which were correct and which were not?



Types of teeth



Humans have three main types of teeth: **incisors**, **canines** and **molars**. Teeth have roots to fix them to your **gums**.

Incisors



Incisors are for biting food. They are straight at the top and sharp.

Canines



Canines are for gripping and tearing food. They are pointed.

Molars



Molars are for chewing and grinding food. They are wide and bumpy.

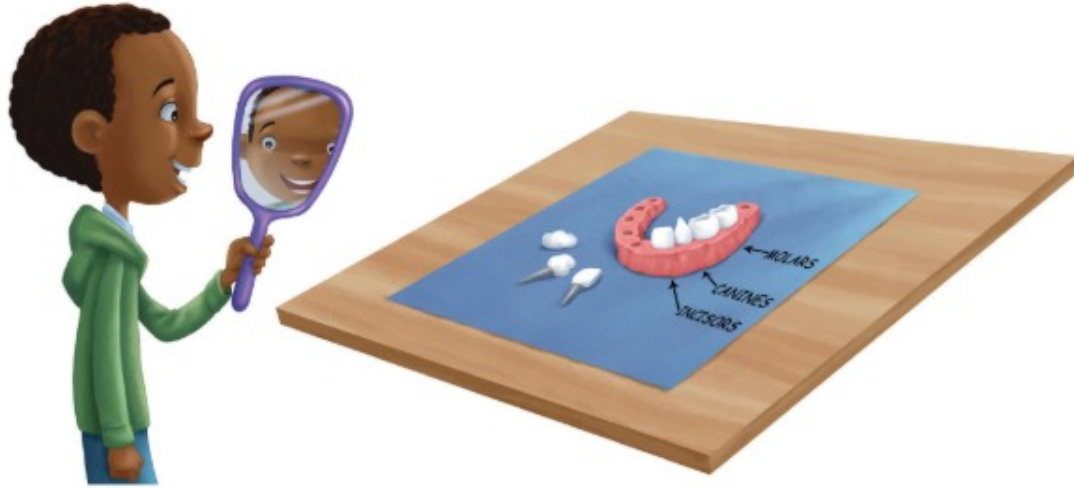
Children have 20 baby teeth. As you grow older these teeth will fall out because larger adult teeth are growing in your gums. Adults have 32 teeth.

Think like a scientist 2

Making model teeth

You will need:
a mirror and some modelling clay

Marcus is making a model of his teeth.
He is using a mirror to see how they look.



Make a model of your own teeth.

Look at them using a mirror. Touch and feel them too.

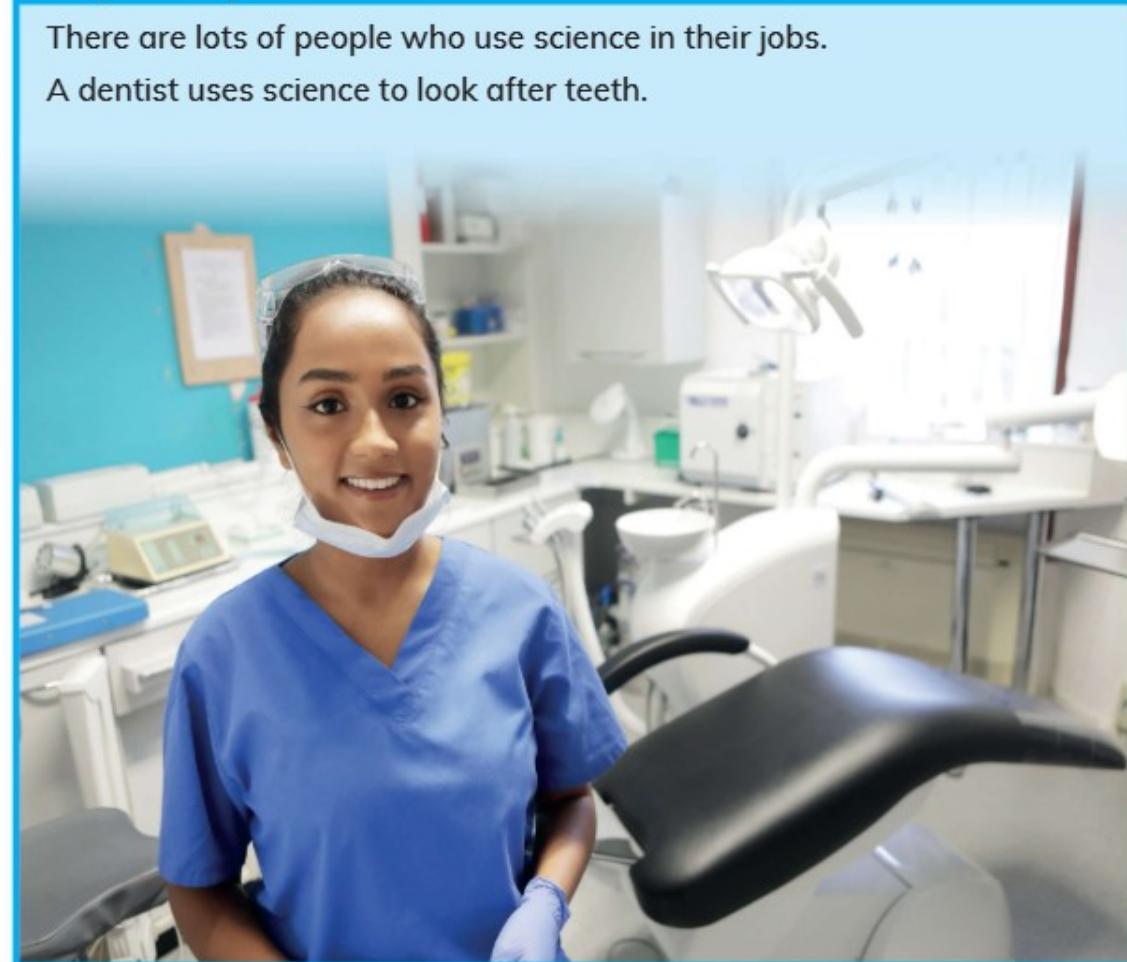
**Wash your hands before and after you touch your teeth.
Do not touch other people's teeth.**

Look what I can do!

- ☐ I can describe the three types of teeth and what they do.
- ☐ I can explain how to look after my teeth.
- ☐ I can say if my predictions are correct or not.
- ☐ I can make model teeth and say how they are like real teeth.
- ☐ I can stay safe when I do an investigation.

Project: People who use science

There are lots of people who use science in their jobs.
A dentist uses science to look after teeth.



Scientists often make and use models.
It is a good way to learn more about something.
What did you learn from making your model teeth?

Continued

A sports coach uses science to plan the best exercises.



The police use science to catch people who have done things that are wrong.



Continued

A zoologist uses science to learn about animals.

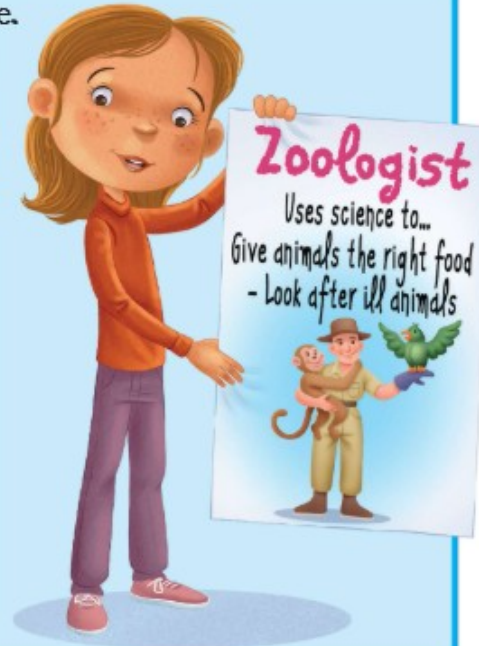


Make a poster about a job that uses science.

You could use books or the internet to find out about one of the jobs shown here.

You could find out about a different job.

Maybe an adult you know uses science in their job.



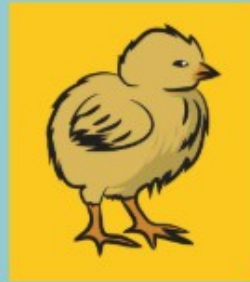
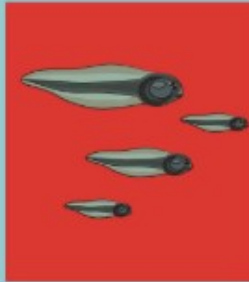
Check your progress

Talk about the answers to these questions with your class.

- 1 How do this newt and this fish look similar?
How do they look different?



- 2 Find the adult animals and their young.



Continued

- 3 Which of these foods are healthy?



- 4 Which of these is exercise?

